ASSESSING MEDIA AUTHENTICITY THROUGH WATERMARKING IN THE CONTEXT OF THE JPEG TRUST STANDARD

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ABSTRACT

This paper presents a watermarking assessment in the context of the emerging JPEG Trust Part 3: Media Asset Watermarking (ISO/IEC 21617-3). The specification complements the JPEG Trust family of standards by enabling a robust content–metadata binding through watermarking, while remaining algorithm-agnostic via explicit signalling of watermark presence, identification of the embedding/extraction technique, and references to external repositories that store Trust Records. It further describes a recommended payload structure and a benchmarking methodology covering imperceptibility and robustness, including modern attacks such as advanced compression (e.g., JPEG XL, JPEG AI) and AI-driven object manipulation. Four state-of-the-art methods from industry and academia (InvisMark, Watermark Anything, TrustMark, and VINE-R) are then evaluated on COCO dataset, highlighting strengths on classical perturbations and persistent vulnerabilities to emerging AI-based attacks, underscoring the need for standard and effective evaluation.

***Index Terms*—** Watermarking standard, JPEG Trust, media authenticity, evaluation.

1. INTRODUCTION

Current technologies applicable to digital media, especially the increasing use of generative AI, make synthetic creation and modification of media assets easy for general users. While there are many benefits, creative outputs and even new knowledge obtained via these technologies, synthetic media artefacts such as images and video also increase the risks of online piracy, cybersecurity fraud, copyright breach, and the spread of mis- and disinformation. The financial costs of inappropriate use of synthetic media assets are at least in the range of US$500 billion per year, and may be as high as US$1 trillion [1].

The JPEG Committee investigation into solutions for this problem led to the 2025 publication of the JPEG Trust Core Foundation (ISO/IEC 21617-1) [2], a framework for establishing trust in digital media assets that relies on metadata-based annotations. However, one of the key challenges is the loss of metadata through routine processing or intentional stripping. In addressing such a challenge, and to facilitate globally interoperable media asset authenticity, JPEG extends the media trust framework described in ISO/IEC 21617-1 with watermarking capabilities, for purposes such as content authenticity, provenance, integrity, intellectual property rights, and labelling through the development of Part 3: Media Asset Watermarking (ISO/IEC 21617-3). Watermarking establishes a link between the metadata and the content, one that is challenging to disrupt without compromising the intended usage of the media asset itself.

This paper describes various components of this developing standard, use cases, the evaluation of selected state-of-the-art algorithms from industry (InvisMark [3] by Microsoft, Watermark Anything [4] by Meta and TrustMark [5] by Adobe) and academia (VINE-R [6]), using the proposed standard, and its significance in the current context. This work marks a significant advancement in the research and commercial development community by providing a standardised framework for watermarking and its evaluation, thereby supporting trusted media asset transactions.

2. STATE-OF-THE-ART

2.1. Watermarking

Watermarking has played a significant role in protecting the integrity of images, such as photographs and digital artworks [6, 7, 8]. The VINE method proposed in [6] adapted SDXL-Turbo by adding a condition adaptor to embed a 100-bit watermark and ConvNeXt-B with a fully connected layer for extraction. In the era of AI, watermarking techniques meaningfully help with tracking AIGC provenance. Recent studies introduced a 'post-hoc' approach, where a watermark can be applied to any image [5, 3, 4]. For instance, TrustMark [5] embeds a 100-bit watermark into any image using an embedder that consists of a pre-processing module, an MUNIT-based network, and a post-processing module, and to decode the watermark, they employ a ResNet-based network as an extractor.

Similarly, InvisMark [3] employs a MUNIT-based encoder and ConvNeXT decoder to encode and decode images with up to 256-bit watermarks. Watermark Anything Model [4] is a deep learning-based method that embeds a 32-bit message using the embedder, consisting of an encoder to compact the image into a latent vector, a learnable lookup table to convert the watermark into embeddings, and a decoder to generate the watermark signal that carries the localised message. We have selected InvisMark [3], Watermark Anything [4], TrustMark [5] and VINE-R [6] for evaluation within the scope of this work.

The popular datasets used for training and evaluating watermarking methods can be grouped into two categories: *1)* real image datasets, including ImageNet [9], COCO [10], and DIV2K [11], and *2)* AI-generated image datasets, including DiffusionDB [12] and SA-1B [13].

2.2. Other Standardization Efforts

In the current landscape, watermarking standards are less about specifying a precise embedding algorithm and more about defining interoperable binding/ identification mechanisms, conformance expectations, and evaluation methodology as relevant to the use case and media type at hand. Evaluation of watermarking in media such as images, video and audio is typically framed around classical criteria, such as imperceptibility, robustness, and payload, plus operational criteria such as security, complexity, real-time, and latency.

A technical report by the Motion Picture Experts Group (MPEG) on MPEG-21 entitled "Evaluation Tools for Persistent Association Technologies" (ISO/IEC TR 21000-11) [14] proposes a framework for watermark/fingerprint robustness assessment against realistic channels and attacks (compression, filtering, resampling, cropping, geometric desynchronization, format conversions, D/A–A/D paths, etc.).

In broadcast and distribution, the Advanced Television Systems Committee (ATSC) A/335 (Video Watermark Emission) [15] is an ATSC 3.0 standard that describes how to embed invisible, robust, and recoverable ancillary data within a video. It specifies a robust payload path through standardization intended to survive transcoding and compression changes and even legacy interface carriage, making conformance-style evaluation naturally revolve around stress testing across codec ladders, transrating, and consumer pipeline impairments.

SMPTE ST 2112-10 [16] is an audio watermarking standard for Open Binding of Content Identifiers (OBID) that embeds persistent IDs (e.g. Ad-ID or EIDR) directly into audio tracks. It implicitly refers to evaluation that includes acoustic/electronic capture conditions (microphone pickup, room impulse responses, loudspeaker distortion) in addition to conventional digital attacks.

The Coalition for Content Provenance and Authenticity (C2PA) [17] accommodates watermarking as an optional soft binding mechanism to embed a resolvable identifier in the media that can be used to look up an active provenance manifest in repositories, but it does not standardize a single watermarking algorithm for this purpose. Instead, it defines guidelines for evaluating soft binding through watermarking for provenance that must be judged not only by signal domain metrics but also by resolution reliability (true/false resolve rates, link persistence under transformations, and ecosystem interoperability) and adversarial conditions (removal, copy/relocation, and laundering).

Existing standards either attempted to address a specific requirements (*e.g.*, [14, 16]) or added watermarking as an additional feature [17] and do not provide a holistic solution in a standardised way. This gap has been addressed in the proposed JPEG Trust Part 3.

3. APPLICATION SCENARIOS

Watermarking can complement metadata- and container-based approaches by enabling information to remain associated with the content itself even when metadata is stripped. This capability enables a wide range of use cases relevant to businesses, governments, and cultural heritage institutions. Common applications include labelling of AI-generated media, Digital Rights Management (DRM), forgery or fraud detection, copyright protection, broadcast signalling, forensic tracing, authenticity and provenance, intellectual property rights (IPR), authorship attribution, and ownership claims. The following subsections discuss the most pertinent use cases within the context of this work in more detail.

3.1. IPR and Digital Rights Management

The works of content creators and rights holders are increasingly exposed to unauthorised copying, redistribution, and attribution omission. In particular, the widespread availability of high-quality digital media and automated content processing pipelines have lowered the barriers for misuse, while complicating the enforcement of intellectual property rights (IPR). More recently, the inclusion of copyrighted works in the training of generative AI models without explicit consent has further intensified concerns among creators, industry stakeholders, and regulatory bodies.

In parallel, rights holders seek mechanisms to retain control over how, where, and under which conditions their works may be accessed, reused, or monetised, forming the basis of Digital Rights Management (DRM). Robust digital watermarking can play a key role in addressing these challenges by enabling the persistent association of rights-related identifiers or messages directly with the media content itself. As a result, even if metadata is stripped, watermarking allows content to remain identifiable, even after common transformations such as compression, cropping, resizing, or transcoding.

As analysed in [18], watermarking-based approaches support a range of IPR- and DRM-related use cases, includ-

ing copyright assertion, content tracking, forensic tracing of unauthorised redistribution, and automated rights enforcement within large-scale distribution ecosystems. When deployed within standardised frameworks, watermarking further enables interoperability between content creators, distributors, platforms, and enforcement mechanisms, thereby reducing fragmentation and supporting consistent rights recognition across heterogeneous media workflows.

3.2. AI-Generated Content Labelling

The rapid proliferation of AI-generated and AI-modified media has raised concerns regarding the misuse of synthetic content. In response, governments and regulatory bodies around the world have introduced legislative measures aimed at improving the identification and labelling of AI-generated media. Within this context, digital watermarking has emerged as a key technical mechanism to support the persistent identification of synthetic content.

A survey of international AI regulation covering 71 countries [19] notes that China was an early proponent of such measures, enacting the Internet Information Service Deep Synthesis Management Regulations in 2022, requiring service providers to apply watermarking or equivalent measures when publishing AI-generated or AI-modified content. Since then, regulatory activity has expanded rapidly: within the EU, the United States, and Canada alone, 129 enacted or imminent legislative instruments addressing AI transparency and labelling have been identified [20]. Examples include the EU AI Act and the California AI Transparency Act, both of which explicitly recognise technical labelling mechanisms for AI-generated media.

From a technical perspective, watermarking offers properties that are well aligned with the requirements of AI content labelling. By embedding machine-detectable information directly into the media asset content, watermarking enables labels to persist even when the content is altered or separated from its original metadata. This makes watermarking particularly suitable for large-scale media ecosystems sensitive to manipulation.

Beyond national legislation, the need for internationally harmonised approaches has been emphasised at the policy level. The 2023 G7 Hiroshima Summit Code of Conduct on AI called for the development of internationally recognised technical standards and the promotion of harmonised solutions for AI transparency and accountability [19].

4. METHODOLOGY

The JPEG Trust Part 3: Media Asset Watermarking framework establishes a link between the media asset and a *Trust Record*, a set of information about the media asset's provenance. The watermarking framework includes aspects of an algorithm-agnostic embedding and extraction process, the signaling of the presence of a watermark, the structure of the watermark, evaluation criteria, and pointers to repositories containing additional information.

4.1. Watermark embedding and detection

As watermarking algorithms may vary in nature and emerge frequently, particularly with new machine learning/AI models, this framework does not standardise any watermarking algorithm but instead provides means to identify the algorithm that is used to embed or extract the watermark. In addition, this framework provides a set of new assertions for inclusion into a Trust Record, which include,

- **Watermark signaling:** Signaling the watermark presence in the media asset is useful for further computation of watermark extraction for information retrieval and labelling the media asset, particularly for AI-generated content.
- **Algorithm:** This framework does not standardise any single watermark embedding or extraction algorithm but provides means to specify information related to the algorithm that was used to embed the watermark and how the extraction could be done. Key information provided includes the name of the algorithm, description of the algorithm, contact information and informational URL for further details of the algorithm.
- **Referencing an external repository:** In the context of the JPEG Trust series of standards, one of the key usages of watermarking is the retrieval of Trust Records and any other relevant information that are securely stored in a repository external to the media asset. This scenario is common when media asset metadata is missing or stripped off. This assertion provides information related to the external repository.

4.2. Watermarking structure

The structure describes a syntax for the *payload* (the data carried by the watermark) within the media asset, indicating the synchronisation pattern, purpose of the watermark, unique identifier (UID) to the media asset, reference to the external repository for metadata retrieval, attribution, time stamp, integrity check, and error correction code. The structure also includes the optional inclusion of user-defined information through unspecified fields. While the payload can be media-agnostic, this standard is optimised for embedding in media asset watermarking within the JPEG ecosystem. An example watermark structure is given in Fig. 1.

Based on the usage scenario, the user may use only mandatory fields or mandatory fields with one or more optional fields. As the payload has an impact on the watermarking imperceptibility and robustness, users may decide what is best suited for their usage scenario.

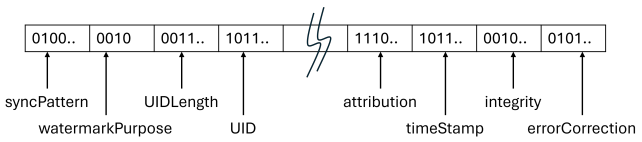


Fig. 1. Watermark structure for a hypothetical use case for illustration purposes only.

4.3. Performance Criteria

Though the algorithms are user-defined, the watermark embedding and extraction processes are expected to be imperceptible (*i.e.*, invisible watermarking) and robust against various processing that the media asset may undergo in the content distribution chain, such as filtering, compression, cropping, or modification. Imperceptibility and robustness are measured against a set of evaluation criteria and can be used to assess the effectiveness of the watermark in a given context using a specified *Watermarking Profile*. The JPEG Trust Part 3 standard provides the framework for such evaluation as defined below.

4.3.1. Embedding Distortion Metric

Imperceptibility refers to a measure of invisible watermark embedding that indicates how similar the original and watermarked media assets are to the human eye. Imperceptibility measures include both objective and subjective measurements. This framework recommends a list of embedding performance metrics measuring imperceptibility and specifies a set of assertions in a Trust Record: *1)* Mean Squared Error (MSE), *2)* Peak Signal-to-Noise Ratio (PSNR), *3)* Weighted PSNR (wPSNR), *4)* Just Noticeable Differences (JND), *5)* Structural Similarity Index (SSIM), *6)* Fréchet inception distance (FID) and *7)* ITU Subjective Quality Measurement.

4.3.2. Robustness against attacks

Watermarking robustness measures how well the extraction algorithms recover the watermark from the host media, which may have undergone several processing steps in the workflow or intentional corruption aimed at destroying the watermark. This framework recommends a list of watermark extraction performance metrics measuring robustness against one or more attacks (listed below) and specifies a set of assertions into a Trust Record: *1)* Normalised Correlation (NC) and *2)* Bit Error Rate (BER).

The existence of the watermark poses risks as the host image may undergo several processing steps during its life cycle. While some processing is common due to the nature of activities such as enhancement or compression, on some occasions, adversarial actors may intentionally destroy the watermark. Together, such processing and intentional attempts to tamper with the watermark are commonly called attacks. Robustness performances are measured using evaluation metrics, such as NC or BER, against these attacks. This framework specifies several such attacks and recommends corresponding assertions into a Trust Record:

- **Signal Processing:** *1)* Gaussian Noise, *2)* Speckle Noise, *3)* Salt & Pepper Noise, *4)* Blurring, *5)* Brightness, *6)* Sharpness, etc.
- **Geometric Processing:** *1)* Rotation, *2)* Resize, *3)* Cropping, *4)* Flipping etc.
- **Removal:** *1)* DCT/Wavelet suppression, *2)* Generative model-based removal, e.g., GAN/diffusion-based.
- **Compression:** *1)* JPEG 1, *2)* JPEG 2000, *3)* JPEG XS, *4)* JPEG XL, *5)* JPEG AI etc.
- **AI-based object manipulation** using an automated / semi-automated prompt-based pipeline (*e.g.*, [21]).

Several of these attacks (e.g., JPEG XS, JPEG XL, JPEG AI, or AI-object manipulations) are newly introduced and better relevant to the current media distribution pipeline.

4.4. Benchmarking Framework

The overall benchmarking framework includes the evaluation criteria described above and allows users to derive a single score for their target application. For example, the user may choose only the last 2 types of attacks for AI-based pipelines and can have an (weighted) average of those two attacks in defining a benchmark for such applications. This is expected to help the algorithm designers to evaluate the strengths and weaknesses of the algorithms in a standardised way.

5. RESULTS AND DISCUSSIONS

To showcase the capability of the proposed framework, we carefully chose four state-of-the-art algorithms ([3, 4, 5, 6]) and evaluated their embedding and robustness performances on example criteria using our evaluation toolkit [1]. In all cases, we have used the original authors' implementation and pretrained models with their default values. As recommended in the original literature, the watermark was created as a series of random binary numbers of 100 bits, except for Watermark Anything [4], where 32 bits watermark was used with the propagation mask to 0.5. We used a subset of 100 images from the COCO dataset for this evaluation, and we averaged the results.

For the newly introduced attacks, we have used *1)* JPEG XL with distance 12 (D=12) and effort 9 and *2)* JPEG AI at quality level 1 (Q=1) using a deep learning-based image compression model [22]. For AI-based object manipulation attacks, we adopted a pipeline described in [21] with two types

[1] https://github.com/JPEG-Trust-Community/watermarking/

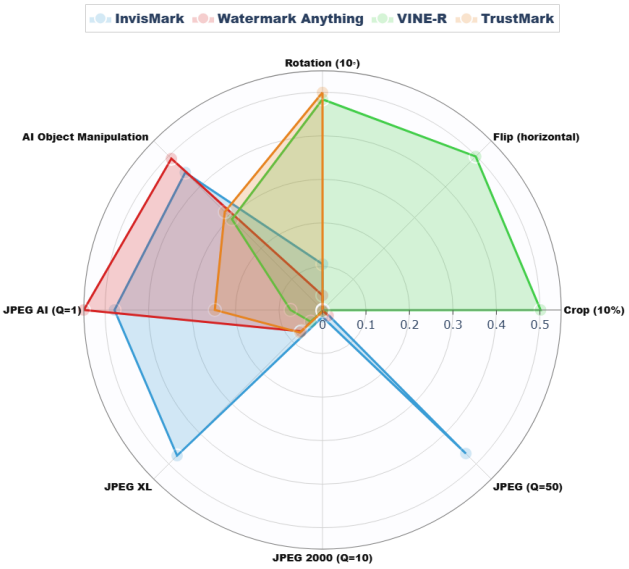


Fig. 2. Robustness performance analysis of selected state-of-the-art methods against select attacks (lower value is better). Generally, most algorithms failed against AI-based attacks, *e.g.*, JPEG-AI and AI object manipulations.

of manipulations: *i)* single manipulations such as removal, replacement, and creation; and *ii)* a combination of more than one type within the watermarked image with manipulations $\geq 25\%$ of the image size.

The embedding distortion and robustness performances are shown in Table 1 and Table 2, respectively. We note that this analysis is not aimed at finding the best algorithm but at showcasing the framework's capability in analysing the strengths and weaknesses of various algorithms. For example, all algorithms have limited capacity (with [4] being the smallest), with reasonable PSNR values but acceptable SSIM or JND values. In robustness analysis (measured in BER), all algorithms show good resilience to standard attacks, but failed against newly introduced AI-based attacks (JPEG AI and AI manipulations), making them unsuitable for some of the emerging usage scenarios. VINE-R [6] shows relatively better performance against compression and AI-related attacks, as it was developed with the focus on generative priors, but it failed in regular geometric attacks. [3] failed against multiple compressions due to its design for high-resolution images, therefore making this algorithm unsuitable for low-bandwidth transmission or devices with small display sizes. An insightful analysis can be observed in Fig. 2. Overall all algorithms indicate poorer performance against AI-based manipulation, which was caused by AI-based models' encoder decoder structures that impact all pixels within the image.

Table 1. Quantitative comparison of image imperceptibility for state-of-the-art watermarking methods on the COCO dataset (subset). **Cap**↑ indicates the watermarking capacity.

Method	**Cap**↑	**SSIM**↑	**PSNR**↑	**wPSNR**↑	**JND**↑	**FID**↓
InvisMark [3]	100	0.9980	49.25	52.59	0.9999	2.1993
Watermark Anything [4]	32	0.9754	34.40	52.17	0.9999	11.3552
VINE-R [6]	100	0.9921	36.29	36.31	0.9916	6.179
TrustMark [5]	100	0.9913	42.34	42.34	0.9965	6.772

Table 2. Robustness comparison (measured in BER) of state-of-the-art watermarking methods on the COCO dataset (subset) against a selection of attacks defined by the JPEG Trust watermarking framework.

Attack	Invis Mark [3]	Watermark Anything [4]	VINE-R [6]	Trust Mark [5]
Blur (5×5)	0.0	0.0	0.0005	0.0002
Brightness (factor 1.3)	0.0023	0.0003	0.0054	0.0142
Sharpness (×1.25)	0.0001	0.0	0.0009	0.0010
Median filter (5×5)	0.0032	0.0	0.0252	0.0007
Rotation (10°)	0.1052	0.0334	0.4837	0.4994
Scale (×0.5)	0.0002	0.0	0.0004	0.0004
Flip (horizontal)	0.0004	0.0003	0.4983	0.0002
Crop (10%)	0.0018	0.0003	0.5016	0.0003
JPEG 1 (Q=50)	0.4663	0.0184	0.0004	0.0042
JPEG 2000 (Q=10)	0.0159	0.00125	0.0021	0.0035
JPEG XL (D=12)	0.4730	0.0693	0.03817	0.0732
JPEG AI (Q=1)	0.4784	0.5490	0.0736	0.2477
AI Object Manipulation	0.4464	0.4915	0.2944	0.318

6. CONCLUSIONS

Digital watermarking, already in use for several decades, has attracted increased attention, particularly for labelling AI-generated/modified content. To address the need for interoperability and a standardised evaluation mechanism, the JPEG committee proposed a new framework for watermarking within the JPEG Trust series of standards (ISO/IEC 21617). This establishes a link between the media-related information, *e.g.*, metadata and the content, one that is challenging to disrupt without compromising the intended usage of the media asset itself. This paper describes the motivation, use cases and various components of this framework. We evaluated four state-of-the-art algorithms and demonstrated the usefulness of the proposed framework in the context of current media distribution usage scenarios. Future work includes incorporation of fragility and security and understanding the impact of payload and various attacks.

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